# Lab 3 – Configuring a Multi-protocol network (OSPF, EIGRP, BGP (eBGP, iBGP))

# Purpose -

Configuring BGP is an important concept to understand and is very important and frequently used knowledge for a future networking professional. This lab would allow students to configure various routing protocols as a review while requiring them to research about BGP to connect each different Autonomous System (AS). This lab was based off of the eBGP lab and requires a deeper and more specialized understanding of a branch of BGP – iBGP.

# Background Information –

This lab contains the usage of three different routing protocols. OSPF, EIGRP, and BGP.

OSPF has been used since 1991, meaning that it has been used for around 35 years now, and it has been widely used and adopted by many enterprises, data centers, and service providers.

A link-state routing protocol that uses a mathematical algorithm to determine the best path to every IP destination in the network. For example, if you have a network with a hundred routers running OSPF, each router knows how to reach every destination in the network via the most optimal path because using OSPF messages, each router advertises practically every detail about every connected link. This results in all routers collectively accumulating information about every device and network in the environment. Subsequently, every router knows how to reach every IP destination via the optimal network path.

OSPF separates the network into different areas, allowing different areas of the network to be controlled for different purposes and at different times. The backbone of this is 0.0.0.0, or Area 0. All other Areas must some sort of connection to Area 0, or an Area Border Router (ABR), which retains different link-state databases for each area it borders.

Within this lab, OSPF was used three times in two different Autonomous Systems (AS).

Enhanced Interior Gateway Routing Protocol (EIGRP) is a Cisco-developed hybrid routing protocol that combines features of both distance vector and link-state protocols. It provides fast convergence, efficient routing updates, and advanced loop-prevention mechanisms. EIGRP employs a composite metric based on bandwidth and delay by default, with the option to include reliability and load.

EIGRP uses the Diffusing Update Algorithm (DUAL) to maintain a topology table of backup routes, enabling rapid failover in case of link failures. Instead of periodic updates, it sends incremental updates only when changes occur, reducing bandwidth usage. It uses multicast (224.0.0.10) for neighbor discovery and authentication to secure routing exchanges. Features such as automatic summarization, route filtering, and unequal-cost load balancing (variance) make it highly scalable and efficient for enterprise networks.

In this lab, configuring EIGRP involves setting up autonomous system numbers, verifying neighbor relationships, and adjusting metric parameters. EIGRP remains to be a widely used IGP within Cisco environments due to its reliability and flexibility.

Both eBGP and iBGP is used within this lab -

External Border Gateway Protocol (eBGP) is an inter-domain routing protocol used to exchange routing information between autonomous systems. It is a core component of internet routing, ensuring efficient communication between different organizations and service providers. Unlike interior gateway protocols (IGPs) like OSPF or EIGRP, eBGP makes routing decisions based on path attributes such as AS Path, Next-Hop, and Local Preference. It prevents routing loops by appending AS numbers to the AS Path attribute and filtering routes accordingly.

eBGP operates over TCP port 179 and requires manually configured neighbor relationships. By default, it assumes peers are directly connected and uses a Time-to-Live (TTL) of 1, but multi-hop BGP can be used for indirect connections. Route filtering, prefix lists, and route maps are commonly used to enforce traffic engineering and security policies, however will not be used in this lab as the different AS(es) are direction Connected.

In this lab, configuring eBGP involves establishing peer relationships. Understanding how BGP propagates and manages routes is key.

Internal Border Gateway Protocol (iBGP) - Internal Border Gateway Protocol (iBGP) is used to exchange routing information within the same autonomous system (AS). iBGP follows the split-horizon rule, routes learned from one iBGP peer are not advertised to another iBGP peer by default. iBGP operates over TCP port 179 and relies on manual neighbor configuration rather than automatic discovery. To ensure full route propagation, what administrators usually use is route reflectors or configure BGP confederations.

In a standard iBGP lab, setting up iBGP involves establishing peer relationships and implementing route reflectors or confederations. Understanding iBGP is critical for designing scalable enterprise and service provider networks. However, some students configured iBGP within a less scalable way and one that will not work within a larger scaled network. They simply set two routers within the same AS to be neighbors and propagated routes through an IGP (OSPF in this case), which isn't realistically scalable and will quickly overload the routers if enough is added, as iBGP is has a default full mesh configuration.

Redistribution of Routes –

Route redistribution allows different routing protocols to share route information, enabling communication between networks running eBGP, EIGRP, and OSPF.  Since these protocols use different metrics, redistribution requires defining appropriate values to ensure compatibility. Without proper filtering, it can lead to routing loops, excessive route propagation, or loss of connectivity. In this lab, students would apply redistribution commands and verifying connectivity to ensure connectivity.

## Lab Summary –

The majority of this lab was setting up the different routing protocols, but the most difficult part was the troubleshooting. A major issue was the redistribution commands, as they were required everywhere for the network to distribute routes across the different AS(es). The base routing of each AS wasn't the difficult part, it was connecting them, as expected, because eBGP was a new concept to the students, and they had to mainly research what kind of commands they needed to configure on each router.

## Lab Commands –

**router bgp** [**autonomous-system-number**]

- Creates and enters configuration mode for a BGP instance with the selected AS number.

**neighbor** [*ip-address*] **remote-as** [*autonomous-system-number*]

- Establishes a BGP neighbor. For iBGP, the remote autonomous system number must match the local AS.

**Neighbor [***ip-address]* activate

**network** [*ip-address*] **mask** [*subnet-mask*]

- Advertises a specific network to other BGP peers. The network must be in the routing table to be advertised.

**ip address** [*ip-address*] [*subnet-mask*]

- Assigns an IP address to an interface.

**address-family ipv4** [mdt | multicast | tunnel | unicast [vrf *vrf-name*] | vrf *vrf-name*]

- Enters an address family configuration mode to configure a routing session using ipv4

**address-family ipv6** [unicast | multicast | vpnv6] [vrf *vrf-name*]

- Enters an address family configuration mode to configure a routing session using ipv6

**router ospf** [*process-id*]

- Enables OSPF with a set process id

**network** [*ip-address*] [*wildcard-mask*] **area** [*area-id*]

- Defines OSPF networks and assigns them to a specific area.

**ip route** [*destination-ip*] [*subnet-mask*] [*next-hop-ip*]

- Creates a static route to ensure the reachability of BGP peer loopback addresses (if not using OSPF or EIGRP).

**show ip bgp summary**

- Displays a summary of BGP peers, including state and established sessions.

**show ip bgp**

- Shows the BGP routing table with learned and advertised routes.

**ping** [*ip-address*] **source loopback0**

- Verifies reachability of the neighbor's loopback address using the local loopback as the source.

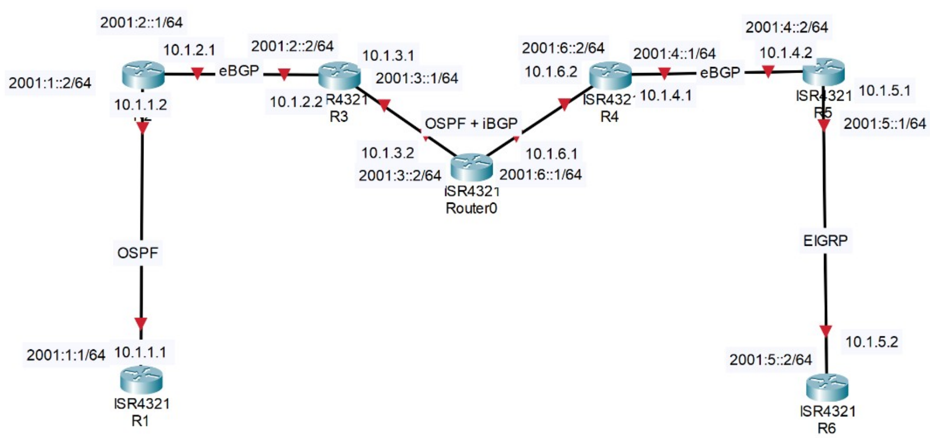
**redistribute** [*protocol*] [autonomous-system-number] [**metric** {*metric value}*] [level]

- Redistributes IPv6 routes from a specified protocol and gives it the specified metric.

**distance** ip-address wildcard-mask [ip-standard-acl | ip-extended-acl | access-list-name]

- Defines an administrative distance for routes that are inserted into the routing table and is used in this lab as a way to show that iBGP is running, as setting iBGP to run at a lower AD would allow iBGP to be shown instead of OSPF.

Network Diagram –




# Lab Configurations –

```
R1 -
Building configuration...
Current configuration : 3875 bytes
Last configuration change at 19:47:53 UTC Fri Jan 24 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
```

```
!
hostname R1
!
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-414644419
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-414644419
revocation-check none
rsakeypair TP-self-signed-414644419
license udi pid ISR4321/K9 sn FDO21080CTW
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
!
interface GigabitEthernet0/0/0
ip address 10.1.1.1 255.255.255.0
negotiation auto
ipv6 address 2001:1::1/64
ipv6 ospf 1 area 0
interface GigabitEthernet0/0/1
ip address 10.1.0.1 255.255.255.0
negotiation auto
ipv6 address 2001::1/64
ipv6 ospf 1 area 0
!
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface Service-Engine0/2/0
```

```
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
router ospf 1
router-id 1.1.1.1
redistribute connected subnets
network 10.1.1.0 0.0.0.255 area 0
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ipv6 router ospf 1
redistribute connected
!
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
Login
End
```

```
IPv4 Route -
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.1.1/32 is directly connected, GigabitEthernet0/0/0
O        10.1.2.0/24 [110/2] via 10.1.1.2, 5d00h,
GigabitEthernet0/0/0
O E2     10.1.3.0/24 [110/1] via 10.1.1.2, 4d00h,
GigabitEthernet0/0/0
O E2     10.1.4.0/24 [110/1] via 10.1.1.2, 4d00h,
GigabitEthernet0/0/0
O E2     10.1.5.0/24 [110/1] via 10.1.1.2, 3d23h,
GigabitEthernet0/0/0
O E2     10.1.6.0/24 [110/1] via 10.1.1.2, 4d00h,
GigabitEthernet0/0/0
```

```
IPv6 Route:
C   2001:1::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:1::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
OE2 2001:2::/64 [110/20]
     via FE80::227:90FF:FED4:F30, GigabitEthernet0/0/0
OE2 2001:3::/64 [110/1]
     via FE80::227:90FF:FED4:F30, GigabitEthernet0/0/0
OE2 2001:4::/64 [110/1]
     via FE80::227:90FF:FED4:F30, GigabitEthernet0/0/0
OE2 2001:5::/64 [110/1]
     via FE80::227:90FF:FED4:F30, GigabitEthernet0/0/0
OE2 2001:6::/64 [110/1]
     via FE80::227:90FF:FED4:F30, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
```

```
R2 -
Building configuration...
Current configuration : 2169 bytes
Last configuration change at 19:45:17 UTC Thu Jan 23 2025
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname R2
!
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
!
ipv6 unicast-routing
!
subscriber templating
```

```
vtp domain cisco
vtp mode transparent
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214328EH
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
!
interface GigabitEthernet0/0/0
ip address 10.1.1.2 255.255.255.0
negotiation auto
ipv6 address 2001:1::2/64
ipv6 ospf 1 area 0
interface GigabitEthernet0/0/1
ip address 10.1.2.1 255.255.255.0
negotiation auto
ipv6 address 2001:2::1/64
!
interface Serial0/1/0
interface Serial0/1/1
interface Service-Engine0/2/0
no ip address
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 2.2.2.2
redistribute connected subnets
redistribute bgp 1 subnets
network 10.1.1.0 0.0.0.255 area 0
network 10.1.2.0 0.0.0.255 area 0
router bgp 1
bgp log-neighbor-changes
neighbor 10.1.2.2 remote-as 2
neighbor 2001:2::2 remote-as 2
address-family ipv4
  network 10.1.2.0
  redistribute connected
  redistribute ospf 1 match internal external 2
  neighbor 10.1.2.2 activate
```

```
  no neighbor 2001:2::2 activate
exit-address-family
address-family ipv6
  redistribute connected
  redistribute ospf 1 metric 10 match internal external 1
external 2
  network 2001:2::/64
  neighbor 2001:2::2 activate
exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ipv6 router ospf 1
router-id 2.2.2.2
redistribute connected
redistribute bgp 1
!
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end

Ipv4 Routes -
Gateway of last resort is not set
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.1.2/32 is directly connected, GigabitEthernet0/0/0
C        10.1.2.0/24 is directly connected, GigabitEthernet0/0/1
L        10.1.2.1/32 is directly connected, GigabitEthernet0/0/1
B        10.1.3.0/24 [20/0] via 10.1.2.2, 4d00h
B        10.1.4.0/24 [20/20] via 10.1.2.2, 4d00h
B        10.1.5.0/24 [20/0] via 10.1.2.2, 4d00h
B        10.1.6.0/24 [20/2] via 10.1.2.2, 4d00h

IPv6 Route:
C   2001:1::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:1::2/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:2::/64 [0/0]
```

```
     via GigabitEthernet0/0/1, directly connected
L   2001:2::1/128 [0/0]
     via GigabitEthernet0/0/1, receive
B   2001:3::/64 [20/0]
     via FE80::B6A8:B9FF:FE47:9471, GigabitEthernet0/0/1
B   2001:4::/64 [20/10]
     via FE80::B6A8:B9FF:FE47:9471, GigabitEthernet0/0/1
B   2001:5::/64 [20/0]
     via FE80::B6A8:B9FF:FE47:9471, GigabitEthernet0/0/1
B   2001:6::/64 [20/2]
     via FE80::B6A8:B9FF:FE47:9471, GigabitEthernet0/0/1
L   FF00::/8 [0/0]
     via Null0, receive
```

```
R3 -
Building configuration...
Current configuration : 2490 bytes
Last configuration change at 19:50:15 UTC Fri Jan 24 2025
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname R3
!
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no logging console
no aaa new-model
no ip domain lookup
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214414TX
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
!
interface GigabitEthernet0/0/0
```

```
ip address 10.1.3.1 255.255.255.0
negotiation auto
ipv6 address 2001:3::1/64
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
ip address 10.1.2.2 255.255.255.0
negotiation auto
ipv6 address 2001:2::2/64
!
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
interface Vlan1
no ip address
!
shutdown
router ospf 1
router-id 3.3.3.3
redistribute connected subnets
redistribute bgp 2 subnets
network 10.1.3.0 0.0.0.255 area 1
router bgp 2
bgp log-neighbor-changes
neighbor 10.1.2.1 remote-as 1
neighbor 10.1.6.2 remote-as 2
neighbor 2001:2::1 remote-as 1
neighbor 2001:6::2 remote-as 2
address-family ipv4
  redistribute connected
  redistribute ospf 1 match internal external 1 external 2
  neighbor 10.1.2.1 activate
```

```
  neighbor 10.1.6.2 activate
  no neighbor 2001:2::1 activate
  no neighbor 2001:6::2 activate
  distance bgp 20 99 1
  distance 99 10.1.6.2 0.0.0.0
exit-address-family
address-family ipv6
  redistribute connected
  redistribute ospf 1 match internal external 1 external 2
  network 2001:2::/64
  neighbor 2001:2::1 activate
  neighbor 2001:6::2 activate
  distance bgp 20 99 1
exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ipv6 router ospf 1
router-id 3.3.3.3
redistribute connected
redistribute bgp 2
!
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end

IPv4 Route:
       10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
B        10.1.1.0/24 [20/0] via 10.1.2.1, 4d00h
C        10.1.2.0/24 is directly connected, GigabitEthernet0/0/1
L        10.1.2.2/32 is directly connected, GigabitEthernet0/0/1
C        10.1.3.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.3.1/32 is directly connected, GigabitEthernet0/0/0
O E2     10.1.4.0/24 [110/20] via 10.1.3.2, 4d00h,
GigabitEthernet0/0/0
B        10.1.5.0/24 [99/0] via 10.1.4.2, 3d23h
O        10.1.6.0/24 [110/2] via 10.1.3.2, 4d00h,
GigabitEthernet0/0/0
```

```
IPv6 Route:
IPv6 Routing Table - default - 9 entries
B   2001:1::/64 [20/0]
     via FE80::227:90FF:FED4:F31, GigabitEthernet0/0/1
C   2001:2::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:2::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:3::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:3::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
OE2 2001:4::/64 [110/10]
     via FE80::267E:12FF:FE55:5721, GigabitEthernet0/0/0
B   2001:5::/64 [99/0]
     via 2001:4::2
O   2001:6::/64 [110/2]
     via FE80::267E:12FF:FE55:5721, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
```

```
R4 - Building configuration...
Current configuration : 3927 bytes
Last configuration change at 14:11:24 UTC Fri Jan 24 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
!
hostname R4
!
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
```

```
crypto pki trustpoint TP-self-signed-1275915594
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1275915594
revocation-check none
rsakeypair TP-self-signed-1275915594
crypto pki certificate chain TP-self-signed-1275915594
certificate self-signed 01
          quit
license udi pid ISR4321/K9 sn FDO214811ZM
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
!
interface GigabitEthernet0/0/0
ip address 10.1.6.1 255.255.255.0
negotiation auto
ipv6 address 2001:6::1/64
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
ip address 10.1.3.2 255.255.255.0
negotiation auto
ipv6 address 2001:3::2/64
ipv6 ospf 1 area 1
!
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
router ospf 1
```

```
network 10.1.3.0 0.0.0.255 area 1
network 10.1.6.0 0.0.0.255 area 1
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ipv6 router ospf 1
Redistribute connected
!
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

```
IPv4 Route:
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O E2     10.1.1.0/24 [110/1] via 10.1.3.1, 4d00h,
GigabitEthernet0/0/1
O E2     10.1.2.0/24 [110/20] via 10.1.3.1, 4d00h,
GigabitEthernet0/0/1
C        10.1.3.0/24 is directly connected, GigabitEthernet0/0/1
L        10.1.3.2/32 is directly connected, GigabitEthernet0/0/1
O E2     10.1.4.0/24 [110/20] via 10.1.6.2, 4d00h,
GigabitEthernet0/0/0
O E2     10.1.5.0/24 [110/1] via 10.1.6.2, 4d00h,
GigabitEthernet0/0/0
C        10.1.6.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.6.1/32 is directly connected, GigabitEthernet0/0/0
```

```
IPv6 Route:
OE2 2001:1::/64 [110/1]
     via FE80::B6A8:B9FF:FE47:9470, GigabitEthernet0/0/1
OE2 2001:2::/64 [110/20]
     via FE80::B6A8:B9FF:FE47:9470, GigabitEthernet0/0/1
```

```
C   2001:3::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:3::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
OE2 2001:4::/64 [110/10]
     via FE80::CE7F:76FF:FE83:16D0, GigabitEthernet0/0/0
OE2 2001:5::/64 [110/10]
     via FE80::CE7F:76FF:FE83:16D0, GigabitEthernet0/0/0
C   2001:6::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:6::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

```
R5 -
Building configuration...
Current configuration : 4846 bytes
! Last configuration change at 19:47:15 UTC Fri Jan 24 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
!
hostname R5
!
boot-start-marker
boot system flash bootflash:isr4300-universalk9.16.09.08.SPA.bin
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
!
ipv6 unicast-routing
!
multilink bundle-name authenticated
enrollment selfsigned
revocation-check none
rsakeypair TP-self-signed-221930004
crypto pki trustpoint TP-self-signed-1275915594
```

```
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1275915594
revocation-check none
rsakeypair TP-self-signed-1275915594
crypto pki certificate chain TP-self-signed-1275915594
license udi pid ISR4321/K9 sn FLM240607T3
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
!
interface GigabitEthernet0/0/0
ip address 10.1.6.2 255.255.255.0
negotiation auto
ipv6 address 2001:6::2/64
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
ip address 10.1.4.1 255.255.255.0
negotiation auto
ipv6 address 2001:4::1/64
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
router ospf 1
router-id 4.4.4.4
redistribute connected subnets
redistribute bgp 2 subnets
network 10.1.6.0 0.0.0.255 area 1
router bgp 2
bgp log-neighbor-changes
neighbor 10.1.3.1 remote-as 2
neighbor 10.1.4.2 remote-as 3
neighbor 2001:3::1 remote-as 2
neighbor 2001:4::2 remote-as 3
address-family ipv4
  network 10.1.4.0
  redistribute connected
  redistribute ospf 1 match internal external 1 external 2
  neighbor 10.1.3.1 activate
  neighbor 10.1.4.2 activate
  no neighbor 2001:3::1 activate
  no neighbor 2001:4::2 activate
```

```
  distance bgp 20 99 1
  distance 99 10.1.3.1 0.0.0.0
exit-address-family
address-family ipv6
  redistribute connected
  redistribute ospf 1 match internal external 1 external 2
  distance bgp 20 99 1
  network 2001:4::/64
  neighbor 2001:3::1 activate
  neighbor 2001:4::2 activate
exit-address-family
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ipv6 router ospf 1
router-id 4.4.4.4
redistribute connected
redistribute bgp 2
!
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end

IPv4 Route:
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
B        10.1.1.0/24 [99/0] via 10.1.2.1, 4d00h
O E2     10.1.2.0/24 [110/20] via 10.1.6.1, 4d00h,
GigabitEthernet0/0/0
O        10.1.3.0/24 [110/2] via 10.1.6.1, 4d00h,
GigabitEthernet0/0/0
C        10.1.4.0/24 is directly connected, GigabitEthernet0/0/1
L        10.1.4.1/32 is directly connected, GigabitEthernet0/0/1
B        10.1.5.0/24 [20/0] via 10.1.4.2, 4d00h
C        10.1.6.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.6.2/32 is directly connected, GigabitEthernet0/0/0
```

```
IPv6 Route:
B   2001:1::/64 [99/0]
     via 2001:2::1
OE2 2001:2::/64 [110/20]
     via FE80::267E:12FF:FE55:5720, GigabitEthernet0/0/0
O   2001:3::/64 [110/2]
     via FE80::267E:12FF:FE55:5720, GigabitEthernet0/0/0
C   2001:4::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:4::1/128 [0/0]
     via GigabitEthernet0/0/1, receive
B   2001:5::/64 [20/0]
     via FE80::521C:B0FF:FE42:AF81, GigabitEthernet0/0/1
C   2001:6::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:6::2/128 [0/0]
     via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

```
R6 -
Building configuration...
Current configuration : 4615 bytes
! Last configuration change at 19:36:01 UTC Fri Jan 24 2025
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname R6
!
boot-start-marker
boot system flash bootflash:isr4300-universalk9.16.09.08.SPA.bin
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
!
ipv6 unicast-routing
!
subscriber templating
```

```
vtp domain cisco
vtp mode transparent
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-2219300048
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-2219300048
revocation-check none
rsakeypair TP-self-signed-2219300048
license udi pid ISR4321/K9 sn FDO214913GF
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
!
interface GigabitEthernet0/0/0
ip address 10.1.5.1 255.255.255.0
negotiation auto
ipv6 address 2001:5::1/64
ipv6 enable
ipv6 eigrp 3
interface GigabitEthernet0/0/1
ip address 10.1.4.2 255.255.255.0
negotiation auto
ipv6 address 2001:4::2/64
!
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
interface Vlan1
no ip address
Shutdown
```

```
!
router eigrp 3
default-metric 10000 100 255 1 1500
network 10.1.5.0 0.0.0.255
redistribute connected
redistribute bgp 3
eigrp router-id 5.5.5.5
router bgp 3
bgp log-neighbor-changes
neighbor 10.1.4.1 remote-as 2
neighbor 2001:4::1 remote-as 2
address-family ipv4
  network 10.1.4.0
  redistribute connected
  redistribute eigrp 3
  neighbor 10.1.4.1 activate
  no neighbor 2001:4::1 activate
exit-address-family
address-family ipv6
  redistribute connected
  redistribute eigrp 3
  network 2001:4::/64
  neighbor 2001:4::1 activate
exit-address-family
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ipv6 router eigrp 3
eigrp router-id 5.5.5.5
redistribute connected
redistribute bgp 3
default-metric 10000 100 255 1 1500
!
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

IPv4 Route:

```
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
B        10.1.1.0/24 [20/0] via 10.1.4.1, 4d00h
B        10.1.2.0/24 [20/20] via 10.1.4.1, 4d00h
B        10.1.3.0/24 [20/2] via 10.1.4.1, 4d00h
C        10.1.4.0/24 is directly connected, GigabitEthernet0/0/1
L        10.1.4.2/32 is directly connected, GigabitEthernet0/0/1
C        10.1.5.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.5.1/32 is directly connected, GigabitEthernet0/0/0
B        10.1.6.0/24 [20/0] via 10.1.4.1, 4d00h


IPv6 Route:
B   2001:1::/64 [20/0]
     via FE80::CE7F:76FF:FE83:16D1, GigabitEthernet0/0/1
B   2001:2::/64 [20/20]
     via FE80::CE7F:76FF:FE83:16D1, GigabitEthernet0/0/1
B   2001:3::/64 [20/2]
     via FE80::CE7F:76FF:FE83:16D1, GigabitEthernet0/0/1
C   2001:4::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:4::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:5::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:5::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
B   2001:6::/64 [20/0]
     via FE80::CE7F:76FF:FE83:16D1, GigabitEthernet0/0/1
L   FF00::/8 [0/0]
     via Null0, receive


R7 -
Building configuration...
Current configuration : 1720 bytes
! Last configuration change at 18:33:40 UTC Mon Jan 27 2025
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname R7
!
boot-start-marker
boot-end-marker
```

```
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
!
ipv6 unicast-routing
!
subscriber templating
vtp domain cisco
vtp mode transparent
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214333H6
license boot level securityk9
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
vlan 10,20
!
interface GigabitEthernet0/0/0
ip address 10.1.5.2 255.255.255.0
negotiation auto
ipv6 address 2001:5::2/64
ipv6 enable
ipv6 eigrp 3
interface GigabitEthernet0/0/1
ip address 10.0.5.1 255.255.255.0
negotiation auto
ipv6 address 2001:5:1::1/64
ipv6 eigrp 3
!
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
interface Vlan1
no ip address
Shutdown
!
router eigrp 3
default-metric 10000 100 255 1 1500
network 10.1.5.0 0.0.0.255
```

```
redistribute connected
eigrp router-id 6.6.6.6
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ipv6 router eigrp 3
eigrp router-id 6.6.6.6
redistribute connected
default-metric 10000 100 255 1 1500
!
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

IPv4 Route:

```
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D EX     10.1.1.0/24 [170/281856] via 10.1.5.1, 4d00h,
GigabitEthernet0/0/0
D EX     10.1.2.0/24 [170/281856] via 10.1.5.1, 4d00h,
GigabitEthernet0/0/0
D EX     10.1.3.0/24 [170/281856] via 10.1.5.1, 4d00h,
GigabitEthernet0/0/0
D EX     10.1.4.0/24 [170/3072] via 10.1.5.1, 4d00h,
GigabitEthernet0/0/0
C        10.1.5.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.5.2/32 is directly connected, GigabitEthernet0/0/0
D EX     10.1.6.0/24 [170/281856] via 10.1.5.1, 4d00h,
GigabitEthernet0/0/0
```

IPv6 Route:

```
EX  2001:1::/64 [170/281856]
     via FE80::521C:B0FF:FE42:AF80, GigabitEthernet0/0/0
EX  2001:2::/64 [170/281856]
     via FE80::521C:B0FF:FE42:AF80, GigabitEthernet0/0/0
EX  2001:3::/64 [170/281856]
```

```
     via FE80::521C:B0FF:FE42:AF80, GigabitEthernet0/0/0
EX  2001:4::/64 [170/281856]
     via FE80::521C:B0FF:FE42:AF80, GigabitEthernet0/0/0
C   2001:5::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:5::2/128 [0/0]
     via GigabitEthernet0/0/0, receive
EX  2001:6::/64 [170/281856]
     via FE80::521C:B0FF:FE42:AF80, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
```

# Issues –

There was still the issue where the OSPF routes were EX2 OSPF routes, which students had to configure the redistribution command of OSPF from "redistribute OSPF 1" to "redistribute ospf 1 match internal 1 external 1 external 2" to distribute all routes. The initial setup of each address family was also quite confusing for the students, as it was again a new concept. Also, the students had set up iBGP in a suboptimal way that makes scalability extremely limited, as iBGP is full mesh by default and would therefore overload routers if there was too many of them, and there would be a significant routing table bloat and potential routing loops. This will become especially apparent when there are complex filtering policies and will quickly overwhelm routers.  However, this is still a way to achieve and show iBGP in small scale networks, and therefore this lab is considered to be complete.

# Conclusion –

This lab provided a great review on previously learned IGPs, OSPF and EIGRP. The students configured seven routers, five had OSPF, four also had BGP, and two had EIGRP. This lab gives them an introduction to iBGP, a popular routing protocol that is frequently used by companies and network administrators which allows them to have an experience similar to a real scenario. Even though it wasn't the standard procedure to set up iBGP, it still works as expansion of knowledge and learn how to set up iBGP with the standard way, with route reflectors (as they complete the writeup), and as the students progressed through the lab, it also seems as through they were becoming more and more confident in their network creation, subnetting, the configuration of previously learned routing protocols, and setting up eBGP.